

10-Day Operating Systems Interview Preparation Plan

Study Method for Every Day

1. Warm-up (5 min): Quick recall questions
2. Build the Concept
3. Why does the OS need this?
4. How does it work internally?
5. Visual intuition and real-world analogies
6. Interview Questions (Easy → Hard)
7. Coding Connection (C++/Java/Linux)
8. Company-Level Questions (Intel, Amazon, Microsoft)
9. Rapid Revision

Day 1 – Processes & Threads

- Introduction: OS functions, User Mode vs Kernel Mode
- Process Basics: Program vs Process, Memory Layout, Life Cycle, Process States
- PCB (Process Control Block)
- Context Switching
- Threads: Process vs Thread, Multithreading, User vs Kernel Threads
- 25 Interview Questions + 5 Scenario Questions

Day 2 – CPU Scheduling

- Scheduling Basics
- Dispatcher
- Scheduling Criteria
- FCFS
- SJF
- SRTF
- Priority Scheduling
- Round Robin
- Multilevel Queue
- Multilevel Feedback Queue
- Starvation
- Aging
- Numerical Problems
- 30 Interview Questions

Day 3 – Synchronization

- Race Condition
- Critical Section
- Critical Section Requirements
- Peterson's Algorithm
- Mutex
- Semaphore
- Binary vs Counting Semaphore
- Monitor
- Spinlock
- Producer-Consumer
- Readers-Writers
- Dining Philosophers
- Interview Questions

Day 4 – Deadlocks

- Resource Allocation
- Four Necessary Conditions
- Resource Allocation Graph
- Safe & Unsafe State
- Banker's Algorithm
- Prevention
- Avoidance
- Detection
- Recovery
- Interview Questions

Day 5 – Memory Management

- Address Binding
- Logical vs Physical Address
- MMU
- Swapping
- Contiguous Allocation
- Fragmentation
- Paging
- Segmentation
- Paging vs Segmentation
- Interview Questions

Day 6 – Virtual Memory

- Virtual Memory
- Demand Paging
- Page Fault
- Copy on Write
- Thrashing
- TLB
- Working Set
- FIFO
- LRU
- Optimal
- Belady's Anomaly
- Second Chance
- Interview Questions

Day 7 – File Systems

- File Concepts
- File Allocation
- Directory Structure
- FAT
- Inodes
- Journaling
- File Permissions
- Hard Link vs Soft Link
- Interview Questions

Day 8 – Disk Management

- Disk Structure
- Disk Scheduling
- FCFS
- SSTF
- SCAN
- C-SCAN
- LOOK
- RAID Basics
- Interview Questions

Day 9 – Linux & System Calls

- System Calls
- fork()
- exec()
- wait()
- Zombie Process
- Orphan Process
- Pipes
- Signals
- IPC
- Shared Memory
- Message Queues
- Socket Basics
- Interview Questions

Day 10 – Final Revision

- Complete Revision
- 100 Rapid-Fire Questions
- 50 MCQs
- 30 Scenario-Based Questions
- Intel-style Mock Interview

Interactive Learning Approach

Every session will be discussion-based rather than lecture-based. We will begin with interview-style questions, build intuition, understand internal working, and finish with rapid-fire interview practice.